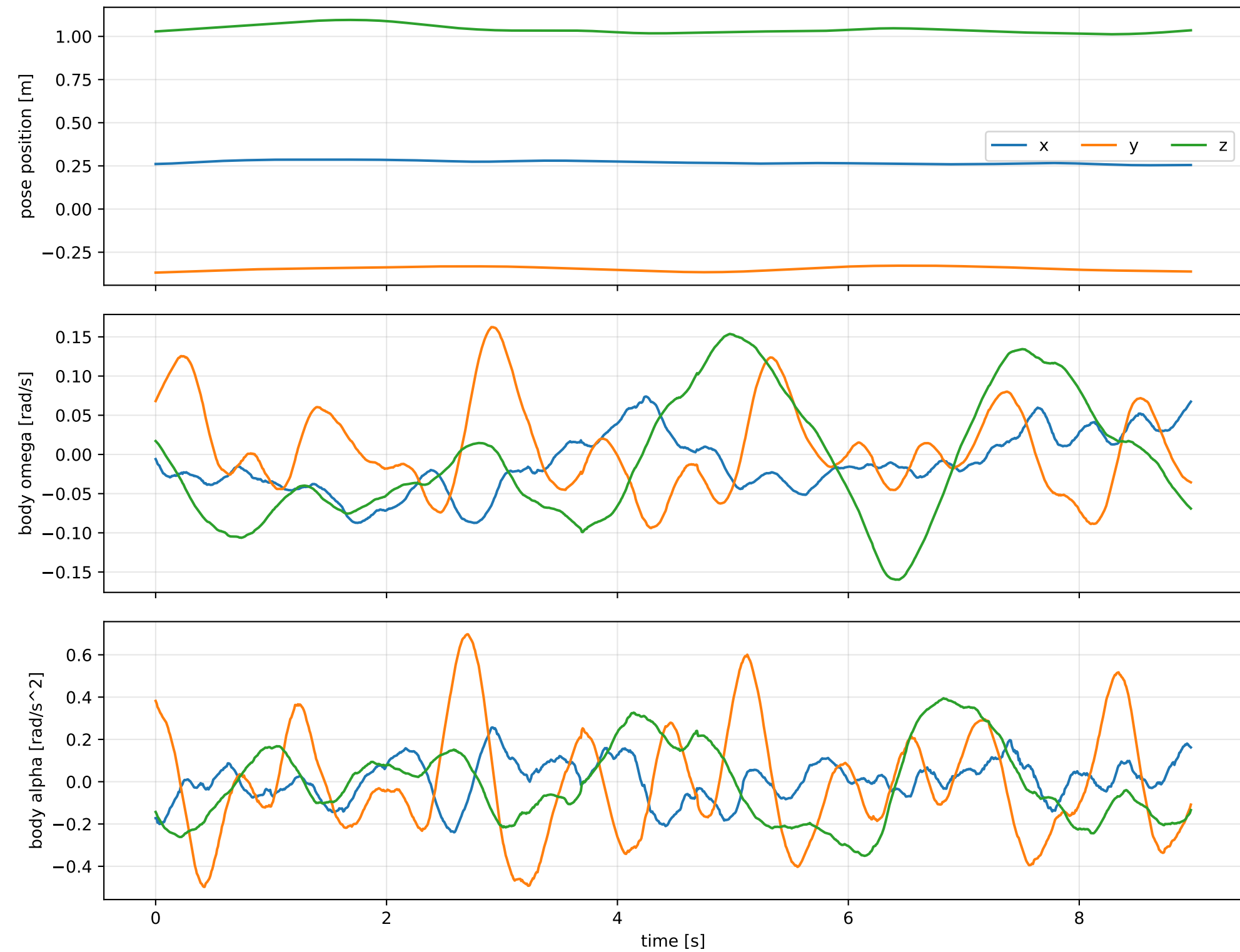
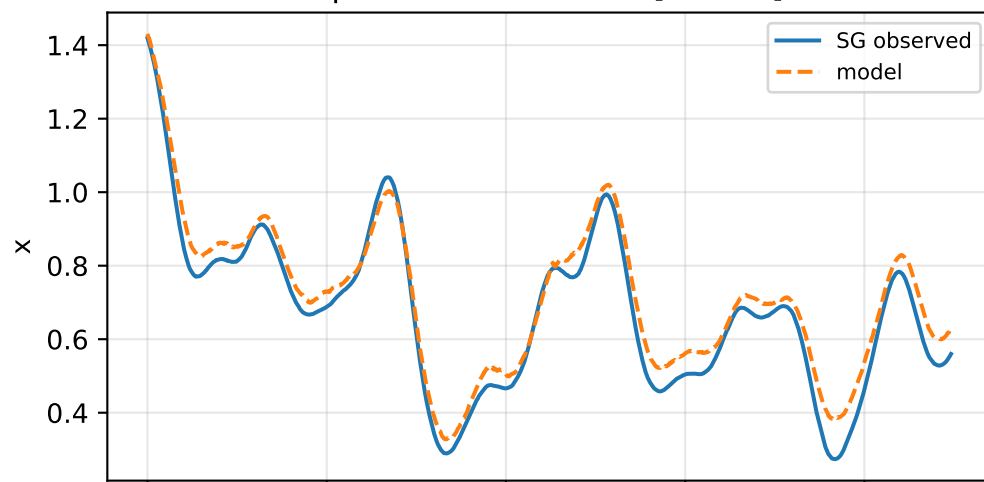


cov_full: SG trajectory and derived motion

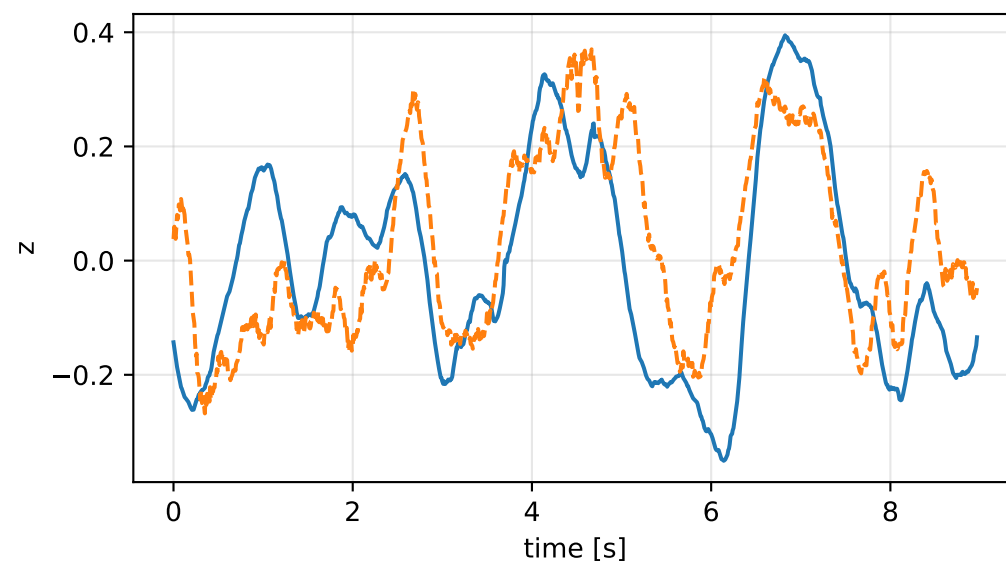
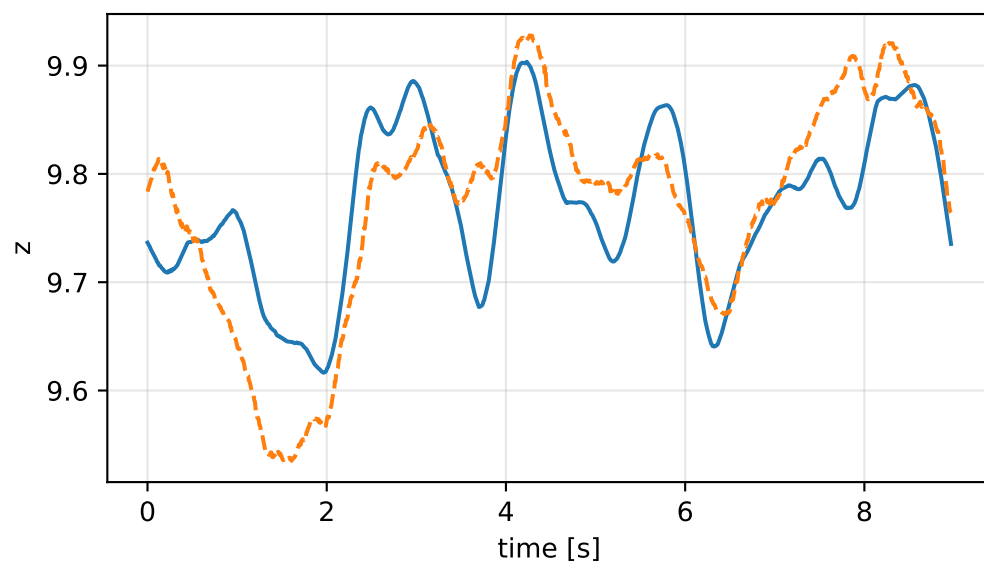
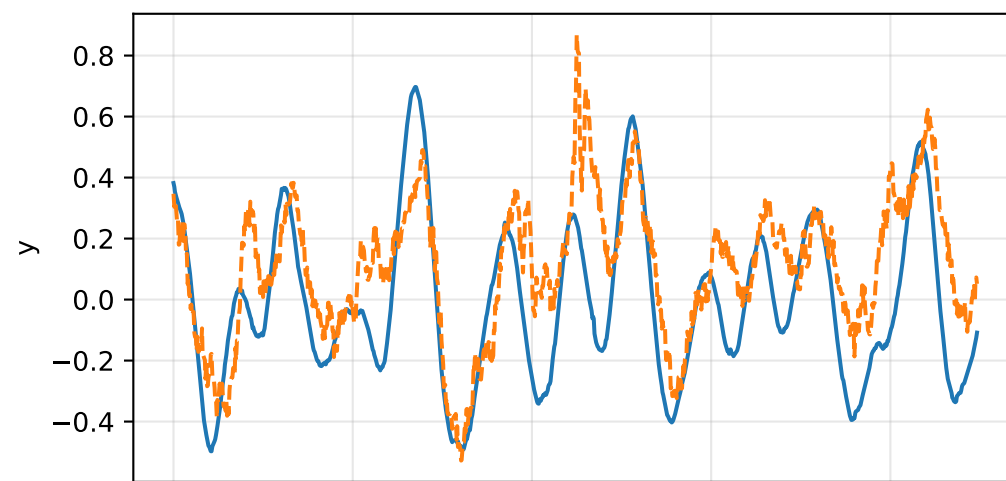
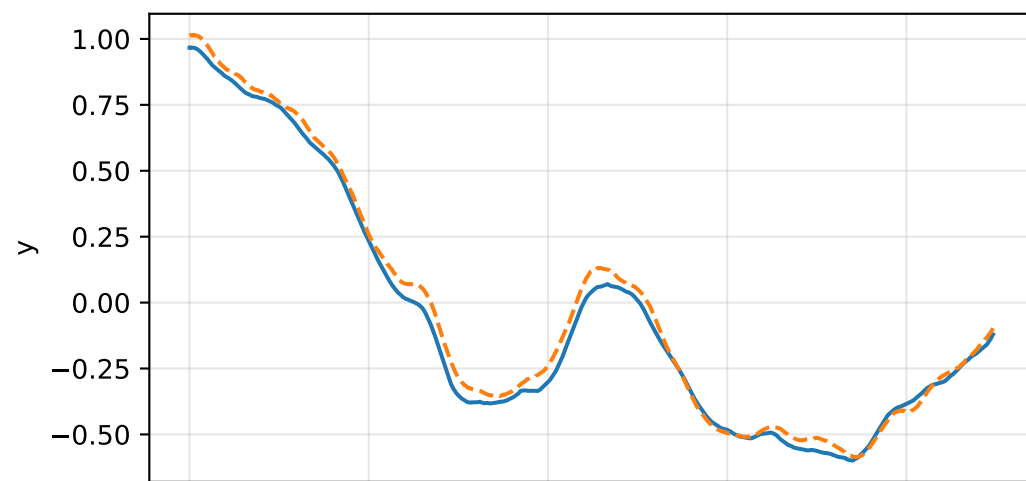
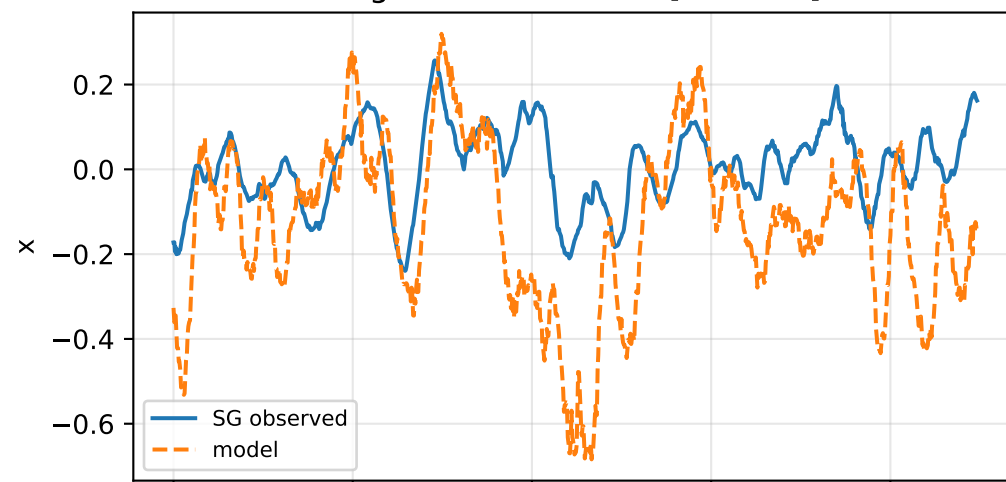


cov_full: acceleration residual objective

specific acceleration [m/s^2]

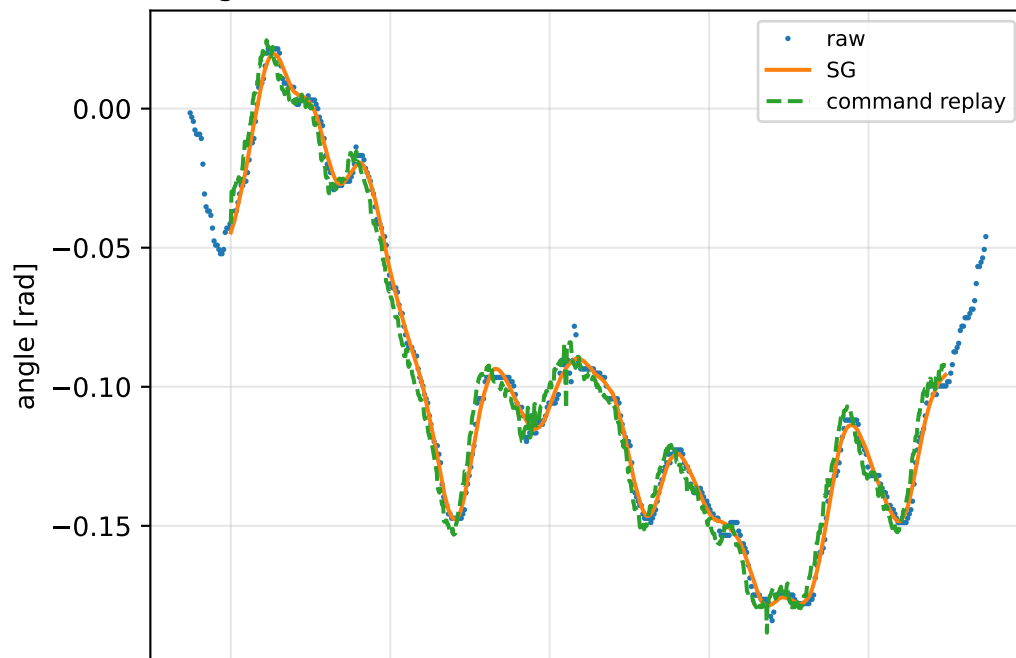


angular acceleration [rad/s^2]

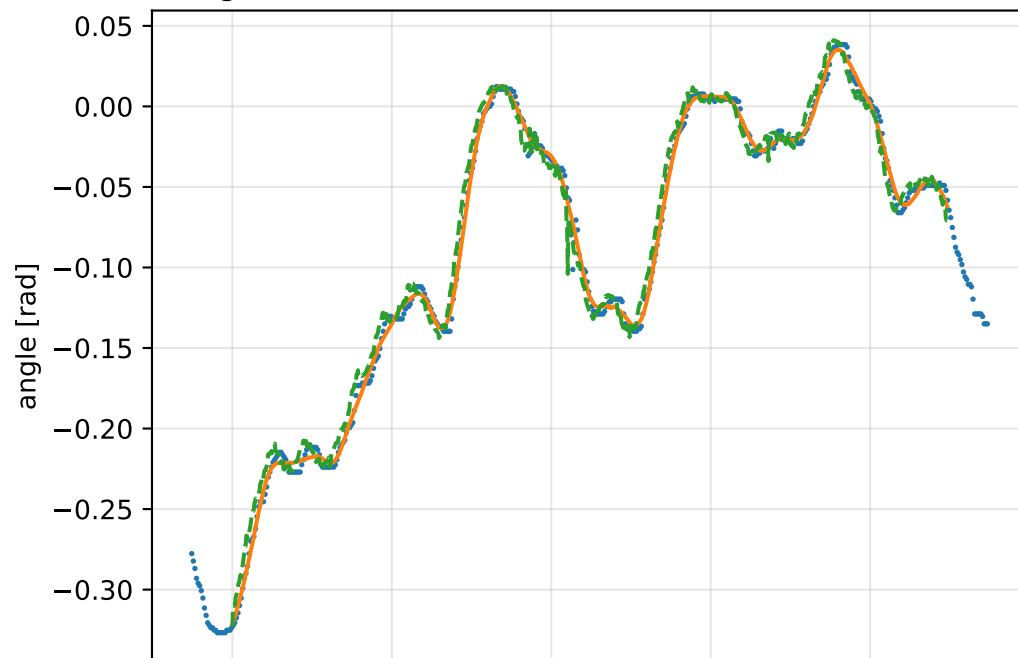


cov_full: gimbal measurement audit (objective=measured_sg)

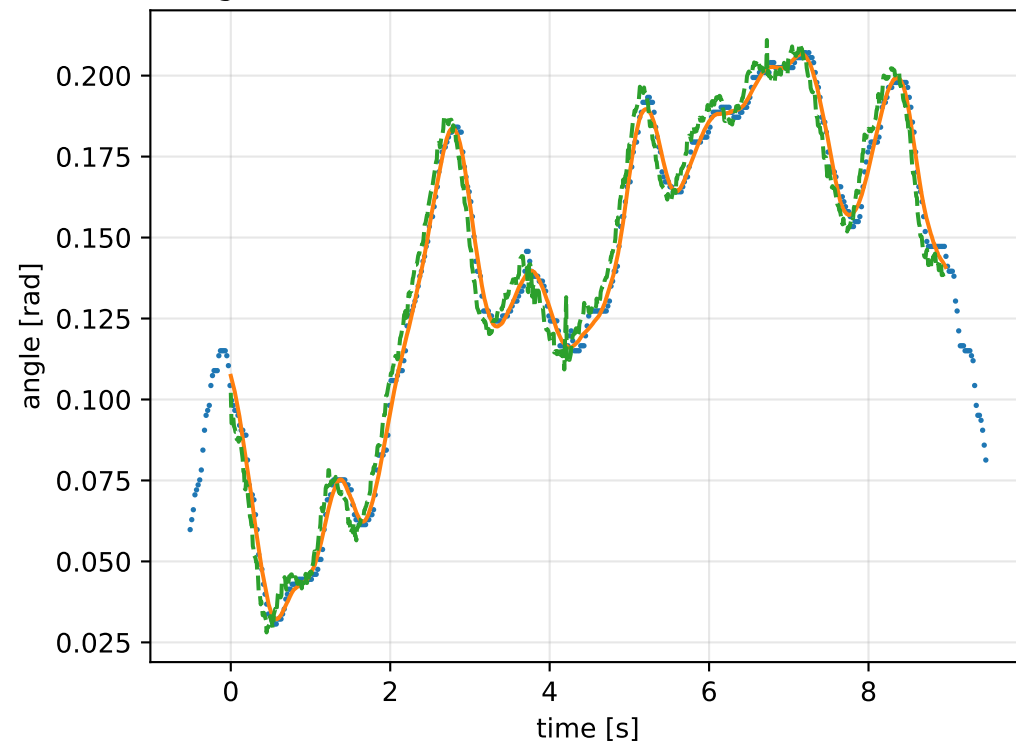
gimbal 1: RMSE=0.006345, max=0.01494 rad



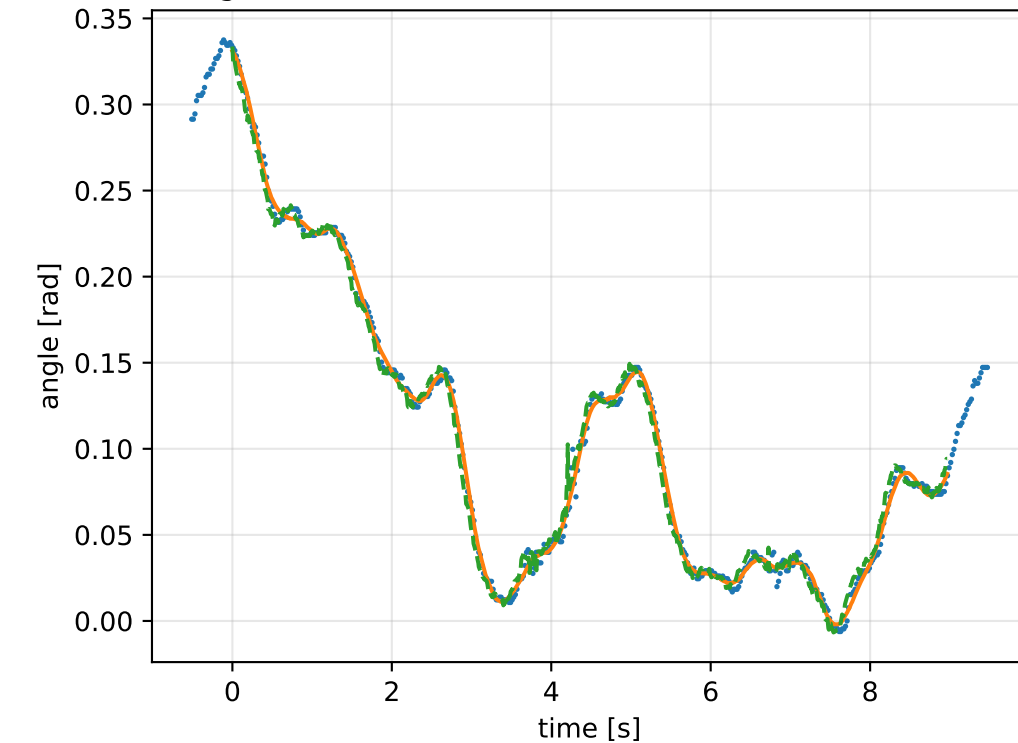
gimbal 2: RMSE=0.008835, max=0.04462 rad



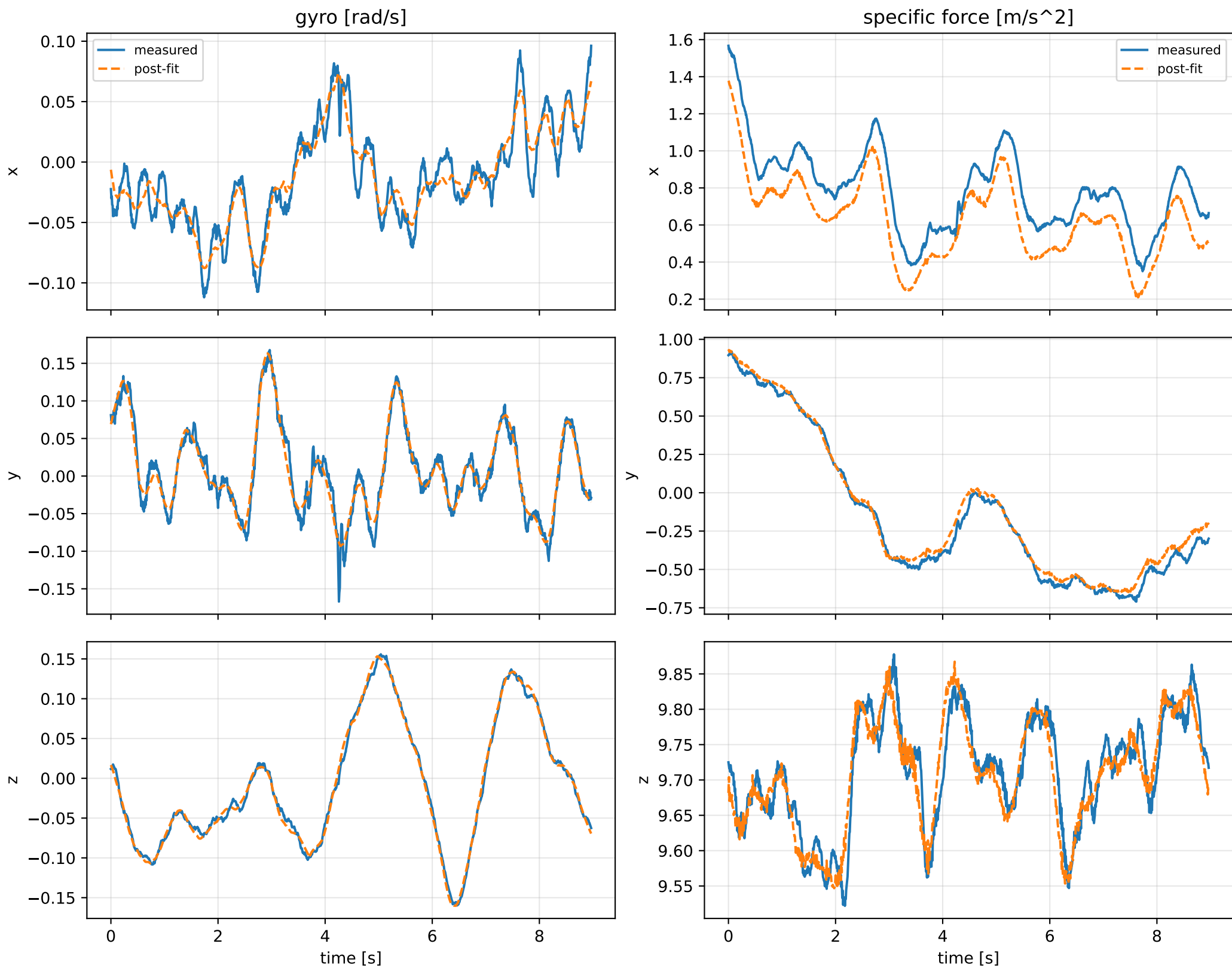
gimbal 3: RMSE=0.007019, max=0.01611 rad



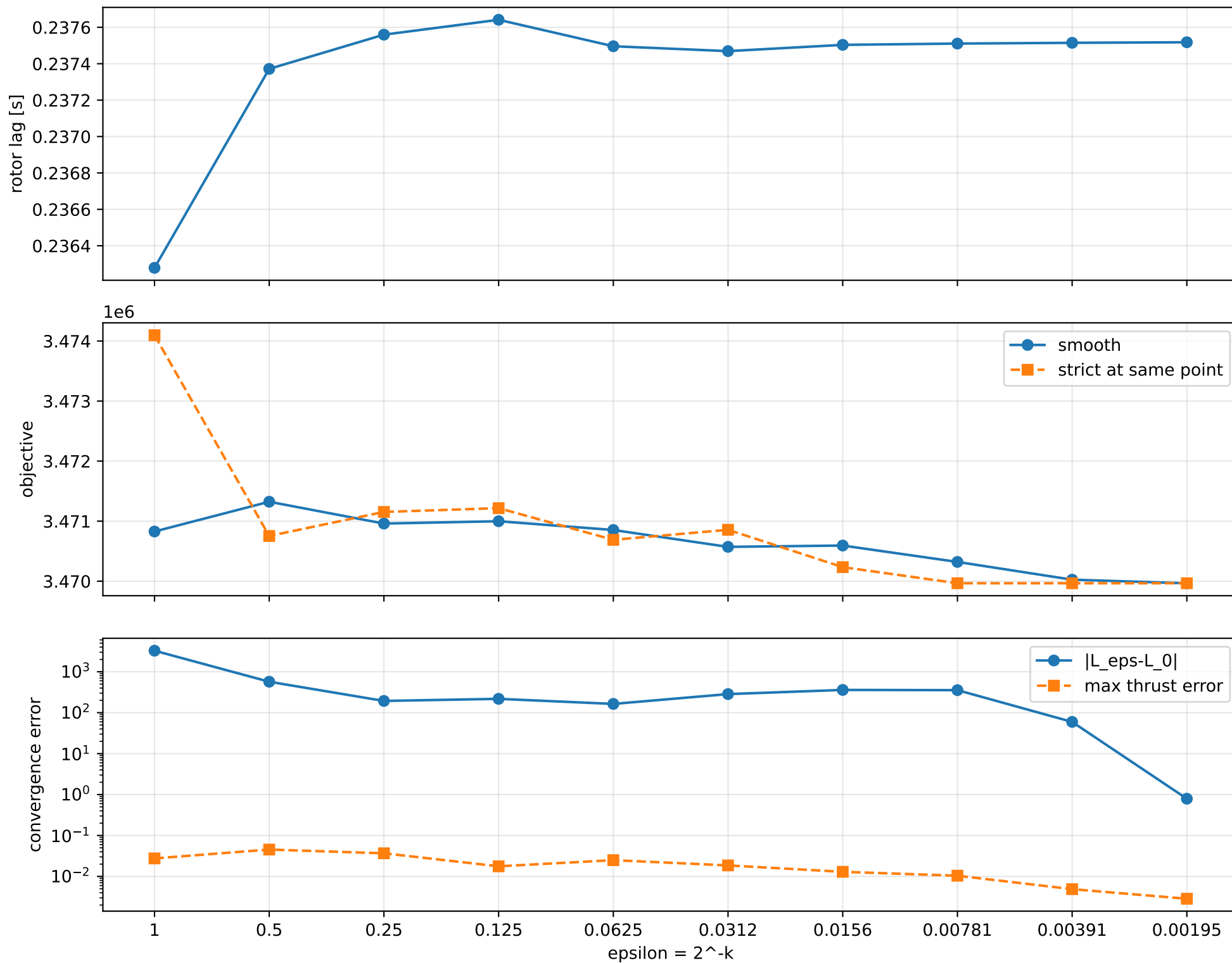
gimbal 4: RMSE=0.006321, max=0.03642 rad



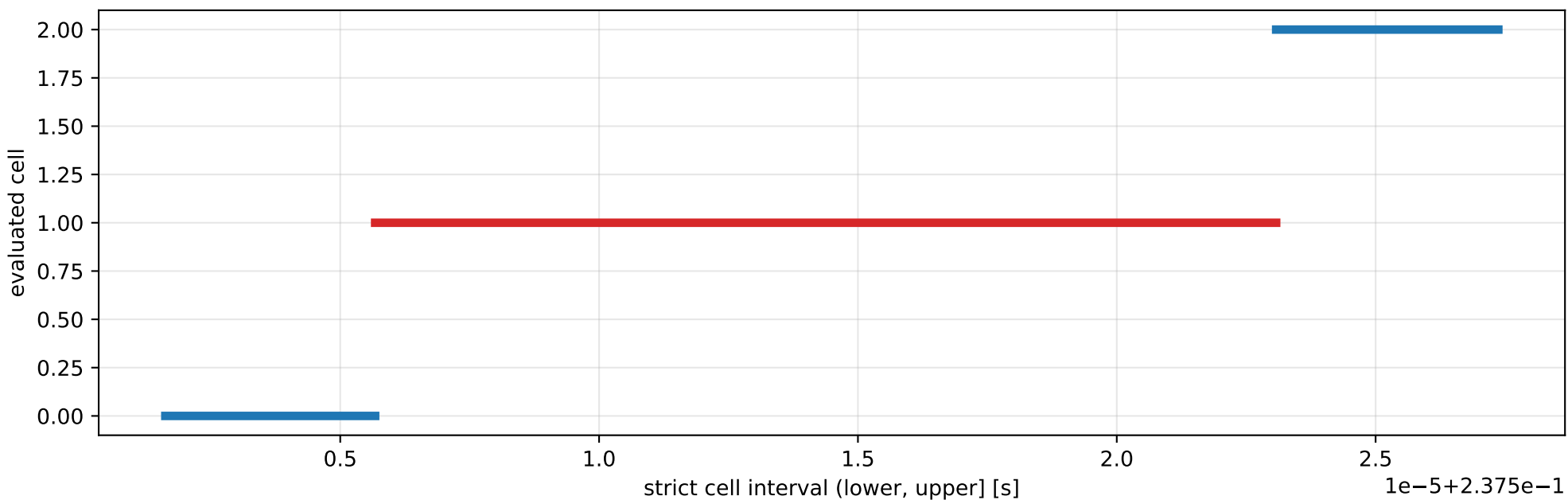
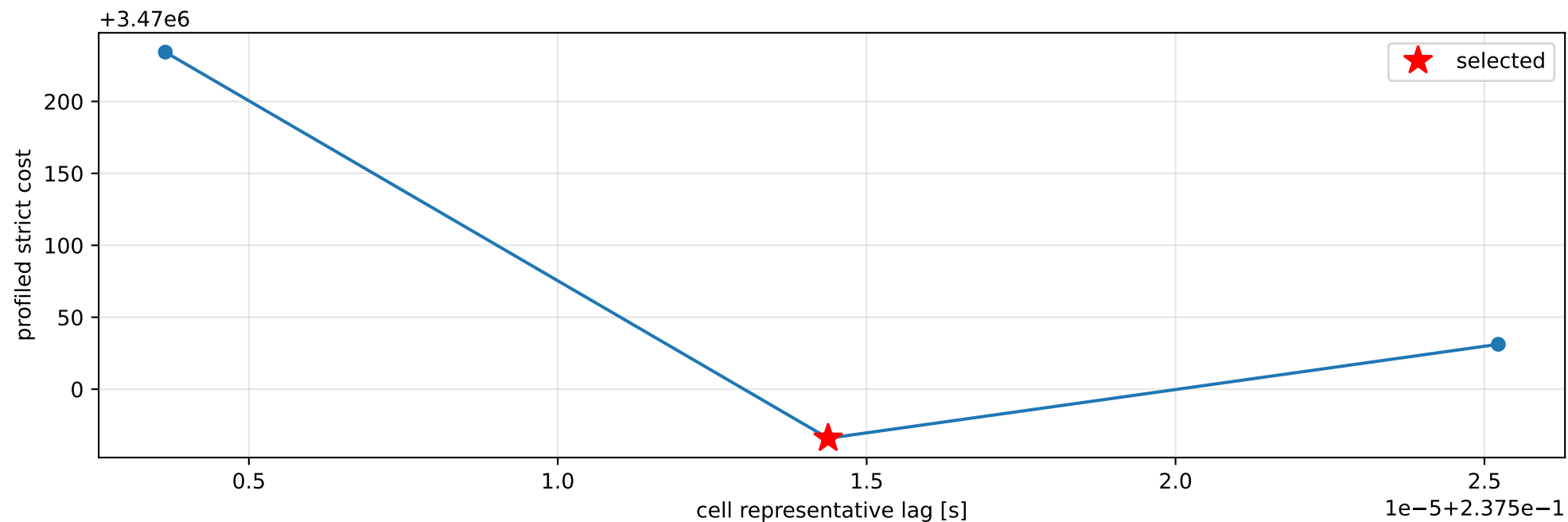
cov_full: IMU comparison (diagnostic only)



cov_full: smooth-to-strict continuation



cov_full: exact strict-ZOH lag cells



selected (0.237505674, 0.237523079] s; final smooth 0.237517854 s; data boundary=False

cov_full: parameter estimate

Case: cov_full

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.89722569

inertia [kg m²]:

[[0.47290395 0.10211167 0.2015896]

[0.10211167 0.47855676 0.03097797]

[0.2015896 0.03097797 0.63031366]]

CoG body [m]: [-0.02370673 -0.0571157 0.02062098]

force effectiveness: [2.2071856 2.14905735 1.00484757 1.43418757]

scale-free J/m [m²]:

[[0.09656568 0.02085092 0.04116404]

[0.02085092 0.09771997 0.00632562]

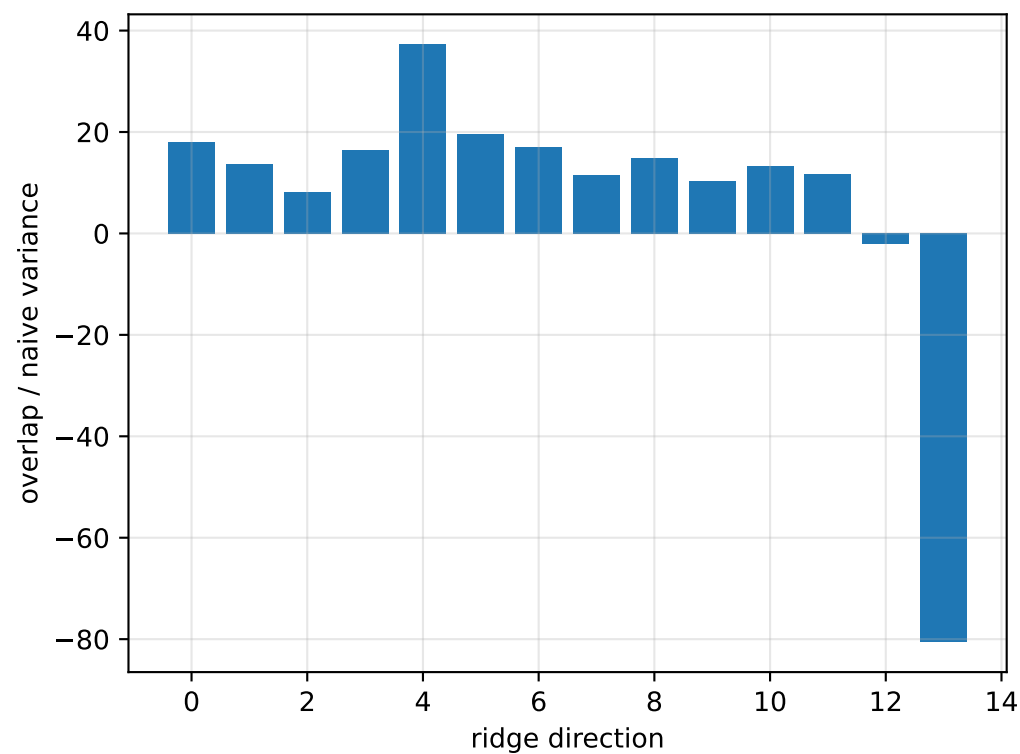
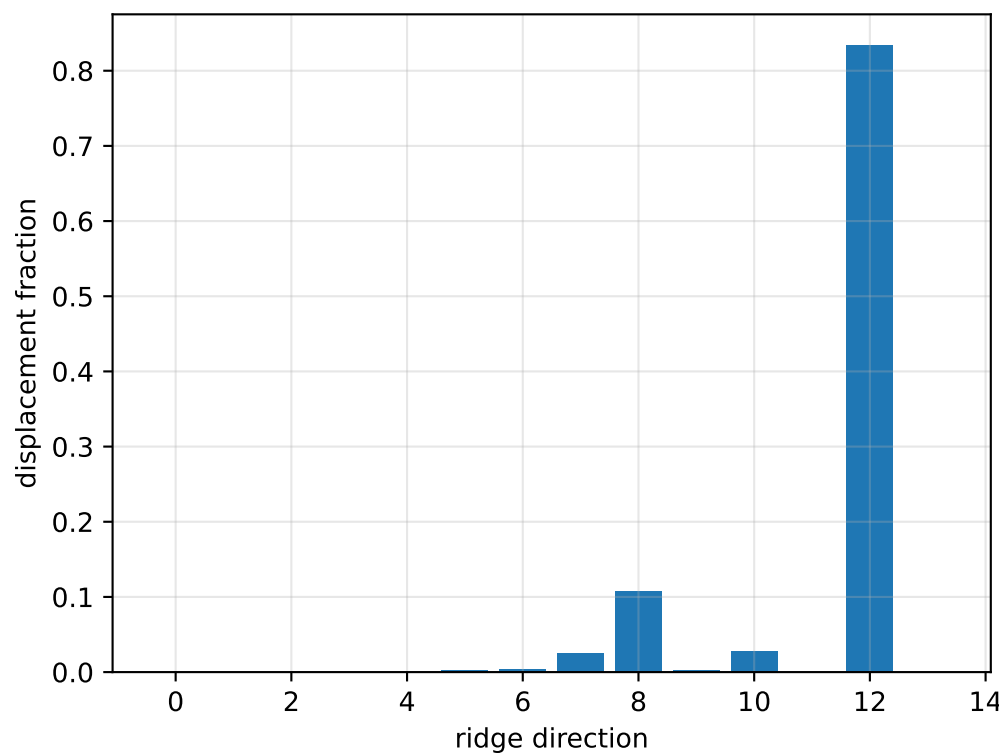
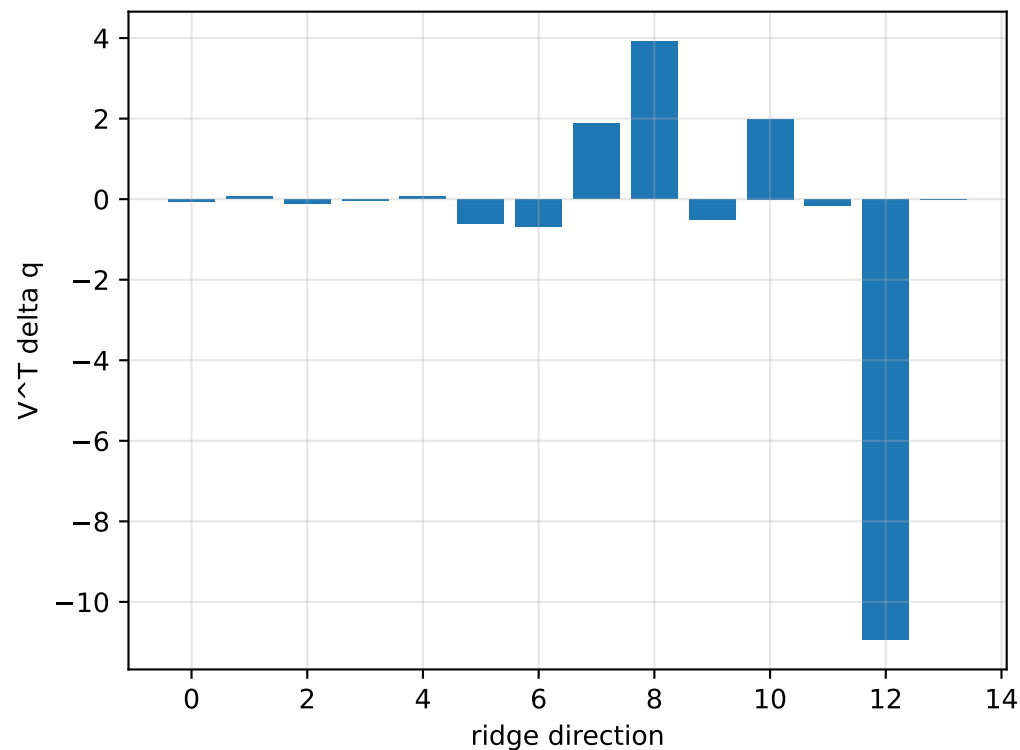
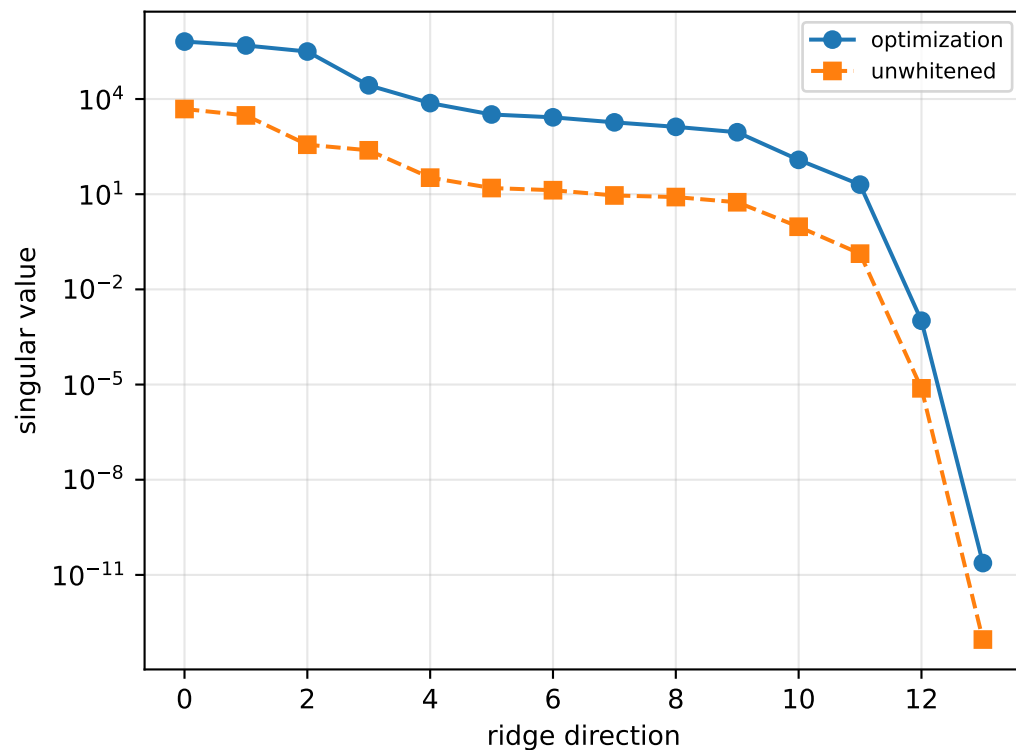
[0.04116404 0.00632562 0.12870831]]

scale-free f/m: [0.45070122 0.43883159 0.20518711 0.29285715]

rotor lag cell: (0.237505674, 0.237523079] s

representative: 0.237514377 s

cov_full: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=5.84\text{e-}11$)



cov_full: optimization vs reference covariance



cov_full: wrench / closure (force max 1.39e-14, torque max 8.33e-17)

